

IDEA SUBMISSION · ROUND 1

The Gen Z Challenge.

A real product-innovation hackathon — from idea to MVP to startup. This is your one-page pitch for the Innovation Round.

TEAM NAME

RealMadrid

COLLEGE

VIT Bhopal University

BRANCH · YEAR

CSE - 4th Year

WHAT CHALLENGE ARE YOU SOLVING?

Problem Statement

INDUSTRY / DOMAIN

Industry - IT

Domain - Cyber Security

TARGET CUSTOMERS

MSMEs often lack the budget and dedicated cybersecurity expertise required for traditional penetration testing. As a customer of AutoPenter, an MSME can affordably and continuously secure its digital assets. AutoPenter provides automated vulnerability scanning and security assessments, empowering small businesses to protect sensitive data, meet compliance, and defend against cyber threats without needing in-house security teams.

PROBLEM DESCRIPTION & CONTEXT

No centralized framework exists that seamlessly integrates all these tools into an automated workflow, reducing testing efficiency, increasing the risk of human error and oversight.

Penetration testing typically requires highly skilled cybersecurity professionals to manually analyze network security, perform vulnerability assessments, and attempt exploitations.

Reports often lack a uniform structure. Some pentesting tools report vulnerabilities but fail to provide detailed mitigation steps, leaving organizations struggling to fix issues.

HOW WOULD YOU SOLVE IT?

Solution 1 / 2

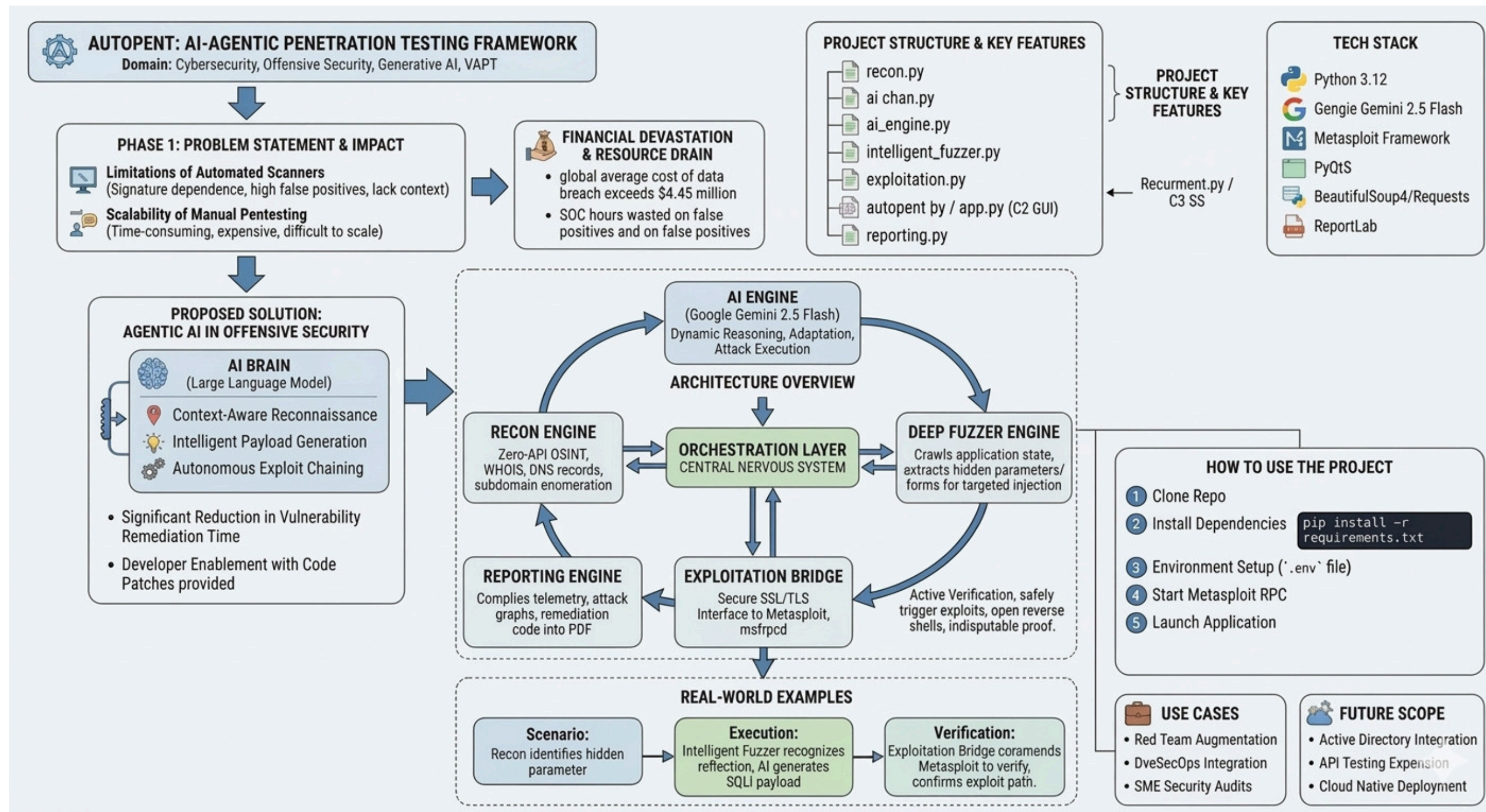
This AI-powered, fully automated penetration testing framework enhances the speed, accuracy, and scalability of security assessments. It modernizes the ethical hacking workflow by overcoming the limitations of slow, manual, and disconnected traditional methods, creating a single, intelligent system for comprehensive security analysis.

It seamlessly combines industry-standard tools like Nmap, Nikto, and SQLMap into one cohesive platform. This is managed through an intuitive Graphical User Interface (GUI) built with Pyqt5 & Flask, providing a simple dashboard for launching tests and reviewing results with ease.

Leveraging AI, the system generates detailed security reports in PDF and text formats. These reports detail identified vulnerabilities, explain their threats, provide base-level remedies, and intelligently prioritize all findings by severity, offering clear, actionable insights for security teams.

HOW WOULD YOU SOLVE IT?

Solution 2 / 2



WHAT TECHNOLOGY POWERS IT?

Solution Components

TECHNOLOGY STACK

- Languages: Python 3.11
- Frameworks: PyQt5, Flask (for backend APIs)
- Libraries:
 - Recon: scapy, sublist3r, shodan
 - Scanning: nmap, openvas_lib
 - Exploitation: msfrpc, pwntools
 - AI/ML: transformers, llama-index, langchain
 - Reporting: reportlab, jinja2, pdfkit
- Databases: SQLite for session & scan data
- AI Model: Gemini for contextual security insights

KEY ASSUMPTIONS

- Automation Reliability: The tool can accurately identify critical vulnerabilities without requiring manual expert analysis.
- User Authorization: Customers have legitimate ownership and legal consent to scan their target digital assets.
- Actionable Remediation: MSMEs will actively use the generated reports to patch and secure their systems.
- System Stability: Target networks and applications can withstand automated penetration testing without experiencing unintended downtime.

HOW DO YOU STAND OUT?

Competitive Analysis

Existing Solutions

Current alternatives include expensive manual penetration testing services, costly commercial vulnerability scanners producing complex reports, and free open-source tools requiring deep technical expertise. Enterprise automated platforms are too pricey and complex for small businesses, while doing nothing leaves MSMEs completely vulnerable until a breach occurs.

Gaps & Limitations

Today's solutions fail MSMEs by being too expensive, overly complex, or both. Manual testing provides only a "point-in-time" snapshot, ignoring continuous threats. Meanwhile, scanners produce hyper-technical reports filled with jargon rather than clear, actionable remediation steps, leaving small businesses overwhelmed, unprotected, and without guidance to actually fix the vulnerabilities.

Your Differentiation

- Continuous vs. Point-in-Time: Instead of expensive, once-a-year manual tests, we provide 24/7 automated assessments that keep up with daily system changes.
- Actionable Clarity over Jargon: We translate complex security vulnerabilities into simple, step-by-step remediation guides so general IT staff or developers—not just cybersecurity experts—can easily fix them.
- Accessible Pricing: Built specifically for MSME budgets, eliminating the prohibitive costs of consulting firms and rigid enterprise security platforms.

THE GEN Z CHALLENGE

Thank you.